

KHASAB PORT, Oman

WMO: 412400

Lat: 26.211N

Lon: 56.244E

Elev: 4

StdP: 101.28

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.8	17.0	-3.4	2.8	32.0	-1.0	3.5	30.4	9.3	21.4	8.3	21.3	2.7	140	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.1	43.0	21.4	41.5	22.0	40.3	22.6	30.4	34.0	29.9	34.1	29.4	34.1	4.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.5	26.4	33.5	28.8	25.3	33.4	28.1	24.3	33.4	102.5	34.0	99.9	33.9	97.3	34.2	32.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.5	6.7	DB	14.2	45.6	1.0	1.0	13.5	46.3	12.9	46.8	12.3	47.4	11.6	48.1	(4)
(5)			WB	7.3	31.4	1.1	0.7	6.5	32.0	5.9	32.4	5.3	32.8	4.5	33.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.2	20.9	22.0	24.6	29.2	33.3	34.7	35.9	35.6	33.9	31.1	26.1	22.1
	DBStd	5.90	1.87	2.21	2.63	3.11	2.67	2.44	1.83	1.76	1.69	1.79	2.06	1.92
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	5	3	1	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6996	338	337	454	577	723	742	802	793	718	653	484	375
	CDD18.3	3958	82	104	196	327	465	492	544	534	468	394	234	117
	CDH23.3	55790	179	379	1462	4189	7282	8007	9155	9008	7500	5802	2340	486
	CDH26.7	35339	7	43	406	2113	4804	5607	6675	6528	5100	3349	673	34
Wind	WSAvg	2.9	2.9	3.0	2.9	2.8	2.9	2.9	3.3	3.5	3.0	2.3	2.5	2.8
Precipitation	PrecAvg	165	39	32	33	6	1	1	1	0	0	3	11	32
	PrecMax	371	188	126	132	46	2	2	18	2	2	30	77	225
	PrecMin	25	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	95	45	33	40	11	1	0	3	1	0	8	19	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.6	29.7	35.4	39.1	44.3	44.9	44.3	43.7	41.7	38.9	33.2	29.2
		MCWB	16.6	17.7	17.0	19.1	20.4	20.1	22.2	21.7	20.9	19.9	20.4	17.5
	2%	DB	26.1	28.0	32.5	37.7	41.4	42.9	42.2	42.1	40.1	37.1	31.7	27.6
		MCWB	16.8	17.8	17.3	19.1	19.8	20.5	23.5	22.6	21.9	20.4	20.7	17.4
	5%	DB	25.2	26.7	30.1	36.0	39.8	40.9	40.5	40.6	38.4	35.6	30.7	26.6
		MCWB	17.3	17.7	17.8	18.7	19.7	21.0	24.6	24.0	22.8	21.5	21.4	17.8
	10%	DB	24.3	25.5	28.6	34.5	38.3	39.1	39.1	39.0	37.1	34.4	29.8	25.8
		MCWB	17.4	17.9	18.1	18.8	20.1	21.8	25.8	25.0	24.3	22.6	21.4	18.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.9	21.9	23.2	26.2	29.1	30.6	31.2	30.7	30.6	28.8	25.1	22.2
		MCDB	24.4	25.6	26.9	29.9	31.6	33.9	33.6	34.6	33.5	32.5	29.3	25.9
	2%	WB	20.1	21.0	22.4	24.9	28.0	29.8	30.6	30.0	29.5	27.9	24.3	21.1
		MCDB	23.9	25.0	26.3	29.0	31.2	33.6	33.9	34.6	33.3	32.5	28.9	25.1
	5%	WB	19.3	20.4	21.7	24.1	27.1	29.2	30.0	29.5	28.7	27.1	23.6	20.2
		MCDB	23.5	24.5	25.8	28.7	31.3	33.3	34.1	34.6	33.4	32.0	28.6	24.5
	10%	WB	18.6	19.7	21.0	23.3	26.4	28.6	29.5	29.0	28.2	26.4	22.9	19.4
		MCDB	23.1	24.0	25.5	28.6	31.4	33.1	34.5	34.5	33.4	31.6	28.4	24.1
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.0	5.3	6.1	6.5	5.8	5.1	5.0	5.1	5.6	5.4	5.7
		MCDBR	6.4	6.2	7.7	8.3	9.7	9.3	7.9	7.8	7.7	7.5	6.4	7.1
		MCWBR	3.9	4.1	4.6	4.6	5.2	6.4	5.5	5.1	5.6	5.8	4.7	4.9
	5% WB	MCDBR	5.3	4.5	5.2	6.1	5.9	5.0	4.3	4.3	4.3	4.7	5.2	5.4
		MCWBR	4.0	3.8	4.3	4.3	5.1	4.6	4.6	4.3	4.3	4.5	4.8	4.6
Clear-Sky Solar Irradiance	taub	0.435	0.497	0.553	0.579	0.581	0.637	0.720	0.676	0.594	0.514	0.449	0.429	
	taud	2.178	1.996	1.858	1.822	1.822	1.748	1.639	1.721	1.842	1.992	2.168	2.197	
	Ebn at Noon	800	775	755	748	745	700	643	669	712	749	780	787	
	Edh at Noon	130	168	203	215	215	230	256	235	203	166	131	123	
All-Sky Solar Radiation	RadAvg	3.96	4.71	5.74	6.71	7.55	7.63	7.06	6.78	6.23	5.42	4.32	3.77	
	RadStd	0.26	0.38	0.40	0.38	0.18	0.17	0.18	0.18	0.11	0.10	0.13	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.46	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	+139	+134

Nomenclature: See separate page